

Tharun Karthik G

Associate MLOps & Backend Engineer

 tharunkarthik2227@gmail.com

 8825933507

 Coimbatore

 [linkedin.com/in/tharun-karthik-g/](https://www.linkedin.com/in/tharun-karthik-g/)

 github.com/thxrun

 leetcode.com/u/Tharun_z

Profile

Emerging MLOps and Python Backend Engineer with a focus on building production-grade machine learning systems. Experienced in developing reproducible pipelines using DVC and MLflow, with a strong foundation in containerization and cloud orchestration on AWS EKS. Adept at automating CI/CD workflows and implementing real-time system observability through Prometheus and Grafana to bridge the gap between data science and scalable software engineering.

Education

Bachelor of Computer Science and Engineering, Sri Shakthi Institute of Engineering and Technology, Coimbatore 7.5 CGPA	2023 – 2027 Coimbatore
HSC Campion Anglo Indian Higher Secondary School 82%	2022 – 2023 Trichy
SSLC Campion Anglo Indian Higher Secondary School	2020 – 2021 Trichy

Projects

Vehicle Insurance Intelligence: End-to-End MLOps on AWS ,
<https://github.com/thxrun/Vehicle-Insurance-Intelligence-End-to-End-MLOps-on-AWS>

- Built a production-grade end-to-end MLOps pipeline for vehicle premium and claim-risk prediction, implementing modular data ingestion, validation, transformation, model training, and S3-based model registry on AWS.
- Containerized the prediction API using Docker and automated CI/CD with GitHub Actions to build, push (ECR), and deploy on AWS EC2, enabling real-time inference and reproducible model updates.

MLOps Infrastructure Capstone: End-to-End Model Lifecycle Orchestration ,
<https://github.com/thxrun/Capstone-Project---Mlops.git>

- End-to-End Orchestration: Engineered a reproducible ML lifecycle by integrating DVC for data versioning and MLflow for experiment tracking, ensuring seamless transition from research to production.
- Scalable Cloud Deployment: Architected a high-availability inference API using Docker and AWS EKS, automating the deployment via CI/CD pipelines and managing artifacts with S3 and ECR.
- Production Reliability: Established proactive system health monitoring through Prometheus and Grafana dashboards, reducing downtime and ensuring real-time performance visibility.

Professional Experience

Machine Learning & MLOps Intern, SmartED Innovations

- Developed optimized supervised and unsupervised models using Scikit-learn, implementing advanced feature engineering and hyperparameter tuning to improve accuracy.
- Engineered reproducible ML pipelines by integrating DVC for data versioning and MLflow for experiment tracking, ensuring consistency across training environments.
- Modernized development workflows by converting research scripts into modular, production-ready Python code to enable scalable model deployment.

Skills

Languages Python,SQL	MLOps MLflow, DVC, Github Actions, Docker, Kubernetes, AWS (IAM,EC2, S3,EKS,ECR), Prometheus & Grafana,Terraform
Backend FastAPI, Object Oriented Programming, PostgreSQL	Git & GitHub Proficient
Linux Proficient	

Certificates

Machine Learning with Python, SmartED Innovations 	Complete Lifecycle of Mlops https://courses.analyticsvidhya.com/certificates/7arqbem2tw 
--	---